BullSO, University of South Florida - Repaired practice edition - Answer key

Division C | Astronomy | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

reasoning: Distance modulus is 5.
d = 10 x 10^(5/5) = 100 pc.
At ten times the distance, flux is one hundredth of its 10 pc value. (3 points).

conclusion: The distance is 100 pc, consistent with a five-magnitude dimming. (3 points).